



Mayank Nagar M.Sc. Data Science

✉ nmayank.790@gmail.com ☎ 015510431014 📍 20179, Hamburg, DE

🔗 mayanknagar10.github.io/my-portfolio 🌐 linkedin.com/in/mayank-nagar10

SUMMARY

Data Science master's student at TUHH with 2+ years of data engineering experience and applied ML research experience in computer vision, generative AI, and scientific image analysis. Skilled in Python, PyTorch, SQL, PySpark, Azure data systems, and ML workflow automation, with hands-on experience in microstructure reconstruction, 3D micro-CT super-resolution, segmentation, and image-quality evaluation.

SKILLS

Programming & Machine Learning — Python, SQL, PyTorch, Scikit-Learn, PySpark, Time-Series Forecasting, Deep Learning, Model Evaluation

LLM & Agentic AI — LangGraph, LangChain, HuggingFace Transformers, RAG, ChromaDB, MCP (Model Context Protocol), Prompt Engineering, LLM Tool Calling

Data Engineering & ETL — ETL, Azure Data Factory, Azure Storage, SQL Server, Kafka, Workflow Optimization

Software & Tools — Docker, Git, Linux, Microsoft Azure, Microservices

Analytics & Visualization — Tableau, Power BI, Excel, Dashboards

EXPERIENCE

Master Thesis, *Helmholtz Centre Hereon GmbH*

04/2026 – Present | Geesthacht, Germany

- Developing **ViTFMDiT** and **ViTDiT** architectures for microstructure reconstruction using diffusion models, flow-matching techniques, and **PyTorch-based** deep learning workflows.
- Building training, validation, checkpointing, and inference pipelines for GPU-accelerated **generative vision models**, with focus on reproducible experimentation and **model evaluation**.
- Designing **latent conditioning and bottleneck** mechanisms to reconstruct microstructure images from compact **semantic representations**.
- Implementing **reconstruction losses** for RGB fidelity, grain-orientation consistency, and boundary preservation, while **monitoring stability** and reconstruction quality.

Data Science Intern, *Harbour Energy*

04/2025 – 09/2025 | Barnstorf, Germany

- Designed ML pipeline with **CycleGAN**, **FFT**, and **interpolation** to upscale **64³ micro-CT scans to 1024³ resolution** for geological analysis.
- Enhanced **image-processing tools** and ML models for **3D super-resolution**, supporting downstream simulation and analytical workflows.
- Automated segmentation, **porosity analysis**, and model-training workflows, **reducing manual intervention** for accurate pore-grain measurements.

Programmer Analyst – Data Engineering

01/2022 – 08/2024 | Pune, India

Cognizant Technology Solutions

- Engineered **ETL pipelines** using **Azure Data Factory**, **Azure Storage**, **SQL Server**, and **PySpark**, reducing data migration latency by 30%.
- Developed and optimized automated data processing workflows, improving **processing efficiency by 25%** and supporting data quality checks.
- Built reports, dashboards, and visualizations using **SQL**, **PySpark**, and **IBM Cognos Analytics** to support stakeholder decision-making.
- Collaborated with **cross-functional teams** and maintained technical documentation, reducing turnaround time by **20%** and onboarding time by **15%**.

EDUCATION

Master of Science in Data Science <i>Hamburg University of Technology (TUHH)</i>	10/2024 – Present Hamburg, Germany
B.Tech. in Computer Science and Engineering <i>Dr. Babasaheb Ambedkar Technological University</i>	08/2018 – 07/2022 Lonere, India

PROJECTS

PlanningCopilot: AI-Augmented Demand Forecasting & Inventory Planning, 	06/2026 – Present
- Built a demand forecasting system with a LangGraph agent over a deterministic Prophet engine, ensuring every forecast traces to a backtested model, never an LLM estimate. - Benchmarked SMA, ETS, Prophet, GBT, and neural baselines on identical holdout windows, achieving 14.4% WAPE and evidence-based model selection. - Engineered a RAG pipeline (ChromaDB) with a custom MCP server and dual-LLM failover (Groq → Gemini), deployed via Streamlit with 31 pytest tests.	
Gen AI based Modeling of RVE for Microstructure Generation	10/2025 – 03/2026
<i>Research Project, TU Hamburg</i> - Designed a ViT-VQGAN for high-fidelity microstructure reconstruction with 260× image compression, addressing the data scarcity challenge in materials science. - Built a novel Dual-Tower SDXL conditioning architecture with 173M trainable parameters across Perceiver Resampler, Pooled Projection, and ControlNet channels. - Compared ViT-VQGAN and SDXL trade-offs, highlighting reconstruction fidelity, compression, diversity, and data augmentation capabilities.	
PRSage: GNN-Powered Code Reviewer Recommendation	11/2025 – 02/2026
<i>System, Deep Learning for Social Analytics, TU Hamburg</i> - Engineered a GraphSAGE-based GNN system to automate reviewer recommendations for GitHub pull requests, reducing review latency in distributed teams. - Constructed developer-PR bipartite graphs from 10,000+ pull requests using the GitHub REST API and CodeBERT embeddings. - Benchmarked link-prediction models using Recall@10 to improve prioritization, matching quality, and recommendation accuracy.	

LANGUAGES

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| • German (B1 Learning) | • English | • Hindi |
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